



JARAMOGI OGINGA ODINGA UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF INFORMATICS AND INNOVATIVE SYSTEMS

DEPARTMENT OF INFORMATION SYSTEMS

**UNIVERSITY EXAMINATION FOR THE DEGREE BACHELOR OF
INFORMATION COMMUNICATION AND TECHNOLOGY
YEAR THREE: SEMESTER ONE**

2ND YEAR 2ND SEMESTER 2024/25 ACADEMIC YEAR

MAIN CAMPUS

COURSE CODE: ITB 1302

COURSE TITLE: NETWORK SYSTEM INTEGRATION AND MAINTENANCE

DATE:

TIME:

INSTRUCTIONS

Answer all Questions

Questions One

6. Define **Network Systems Integration** and explain its importance in modern IT infrastructure.
7. Describe three key **network topologies** and their advantages in system design.

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8. Explain the role of **Physical Layer protocols** in supporting communication across data networks.
 9. What key steps are involved in **configuring and implementing** a new network system?
 10. Explain the difference between **Network File System (NFS)** and **Network Information Services (NIS)**.
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Questions Two

11. Your company is planning to integrate Linux, Mac OSX, and Windows systems into a single network. Explain the steps you would take to ensure a seamless **network integration**. (10 marks)
12. You are assigned to configure a **Samba server** for file and printer sharing in a small office. List and explain the key configurations needed. (10 marks)
13. A company is experiencing slow network performance. As a **network administrator**, outline the key steps to diagnose and resolve the issue. (10 marks)

Questions Three

14. **System Maintenance and Security**
 - a) Discuss the significance of **network monitoring** in ensuring system security.
 - b) Explain how **system audits** can help optimize network performance. (20 marks)
15. **Project Management in Network Integration**
 - a) Outline the key considerations when planning a **network integration project**.
 - b) Discuss the importance of **documentation** in network system maintenance. (20 marks)

Questions Five

1. **System Analysis and Design**
 - o Explain the concept of **system analysis and design** in the context of network systems.
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- Describe the steps involved in gathering customer requirements for a new network system.

2. Network Administration

- Define **network administration** and explain its role in maintaining a stable IT infrastructure.
- Identify and explain three key responsibilities of a network administrator.

3. LDAP and Network Security

- What is **LDAP**, and how is it used in network authentication?
- Discuss two advantages of using LDAP for identity management in an enterprise network.

4. File Sharing Protocols

- Explain the role of **Samba** in a network environment.
- Compare **Samba** with **FTP** and describe a scenario where each would be used.

SULTAN PALACE DEVELOPMENT LIMITED

Description	Previous meter reading	Previous reading date	Current meter reading	Current reading date	Actual consumption (kw/m³)	Unit price (Ksh)	Total Price (Ksh)
Electricity Bill	12443	01-Nov-25	12684	30-Nov-25	241	36	8676
Water Bill	409	01-Nov-25	410	30-Nov-25	1	205	205
service Charge		01-Nov-25		30-Nov-25	1	15000	15000
						Total	23881

Billing Statement

Statement #: CB2-202

Handover Date: 3/12/2019

Bill To: [Mr. Peter M. Ng'ang'a & Mrs. Rose N. No'ano'a
CB2-202

Sultan Palace BR,

Kikambala,

Mombasa, Kenya

Remittance Amount Enclosed: KES 277,360.00

COMMENCING DATE	ENDING DATE	Invoice #	Description	Amount	Payment	Balance
3/12/2019	31/12/2019	BAL B/F	WATER, POWER & SERVICE	253,479.00		253,479.00
1/11/2025	30/11/2025	WE-CB2-202-1	WATER, POWER & SERVICE	23,881.00		23,881.00
Total						KES 277,360.00

Reminder: Please include the invoice number on your check.

Terms: Balance due in 15 days.

REMITTANCE		
Customer Name:	[Mr. Peter M. Ng'ang'a & Mrs.	Account Details:
Customer ID:		Account Name: Sultan Palace Development LTD
Statement #:	CB2-202	Account Number: 0104023997400
Date:	01/12/2025	Bank: Standard Chartered Bank Kenya LTD
Amount Due:	KES 277,360.00	Branch Details: Chiromo Branch
Amount Enclosed:	KES 277,360.00	Branch Code: 02-084

SULTAN PALACE DEVELOPMENT LIMITED

Description	Previous meter reading	Previous reading date	Current meter reading	Current reading date	Actual consumption (kw/m ³)	Unit price (Ksh)	Total Price (Ksh)
Electricity Bill	8743	01-Nov-25	8745	30-Nov-25	2	36	72
Water Bill	1034	01-Nov-25	1043	30-Nov-25	9	205	1845
service Charge		01-Nov-25		30-Nov-25	1	20000	20000
						Total	21917

Billing Statement

Statement #: BH-A901

Handover Date: 17/12/2019

Bill To: [Mr. Peter M. Ng'ang'a & Mrs. Rose N. Na'ana'a
BH-A901

Sultan Palace BR,

Kikambala,

Mombasa, Kenya

Remittance Amount Enclosed: KES 277,220.00

COMMENCING DATE	ENDING DATE	Invoice #	Description	Amount	Payment	Balance
17/12/2019	31/12/2019	BAL B/F	WATER, POWER & SERVICE	255,303.00		255,303.00
1/11/2025	30/11/2025	WE-TA9-001-1	WATER, POWER & SERVICE	21,917.00		21,917.00
Total						KES 277,220.00

Reminder: Please include the invoice number on your check.

Terms: Balance due in 15 days.

REMITTANCE		
Customer Name:	[Mr. Peter M. Ng'ang'a & Mrs. Na'ana'a]	Account Details:
Customer ID:		Account Name: Sultan Palace Development LTD
Statement #:	BH-A901	Account Number: 0104023997400
Date:	01/12/2025	Bank: Standard Chartered Bank Kenya LTD
Amount Due:	KES 277,220.00	Branch Details: Chiromo Branch
Amount Enclosed:	KES 277,220.00	Branch Code: 02-084